expected to be somewhere near the same figure as given above, but from a 90-lb. brake pipe pressure a tight brake cylinder or a brake system free from leakage with 8 inches piston travel would accumulate about 65 lbs. pressure in the brake cylinder as a maximum and be subject thereafter to the same results from leakage that would be encountered with 110 lbs. brake pipe pressure. You will understand that the 20-lb. brake pipe reduction from either 110 or 90 lbs. would result in about 50 lbs. brake cylinder pressure, and if thereafter the leakage was at no greater rate than you mention there might be no further movement of the triple valve, and brake cylinder pressure would reach no higher figure, or there might be a movement during which the auxiliary reservoir and brake pipe pressure would reduce equally, but in which brake cylinder pressure could not increase as a result of leakage. (d) The advantage, if any, would be offset by other conditions; 110 lbs in the auxiliary reservoir in freight service would increase the volume of free air in the reservoir and provide a greater margin of safety in grade operation, but if all cars were not heavily loaded it would produce wheel sliding and likely other troubles: hence, 90 lbs. brakepipe pressure is the maximum recommened for freight service. Where the requirements of service exceed the capacity of the standard freight brake operated with 90 lbs. brake pipe pressure, the needed improvement should be met with an adequate brake system rather than by means of an increase in pressure. Such equipments known as the "Empty and Load" brake have been perfected and are in service.

Brake Sticking.

J. T., Baltimore, Md., writes: I had an 8-car passenger train and after an application one brake on the sixth car failed to release; the triple valve was tested and found to pass all tests on the rack, and the verdict was that I was responsible on account of having made too light a brake pipe reduction. Will you please explain to me how a good triple valve will fail to release after a light brake pipe reduction?

A.—Provided that you have not overcharged the brake pipe and auxiliary reservoirs and have a brake pipe feed valve in first class condition, you cannot make a brake pipe reduction that will apply all of the brakes and still be so light as to prevent the release of any one of eight perfect triple valves. The reason that the light reduction causes stuck brakes is that triple valves will get into such a condition that they will fail to release after a very light brake pipe reduction, but will release after a 10 or 12-lb. brake reduction, as in the latter case there is a greater difference between the pressure in the brake pipe and auxiliary reservoirs and the adjustment of the feed valve, and the capacity of the feed valve varies with any difference in the pressure it is working against. You should have called your accuser's attention to the fact that a triple valve test in the shops at this time of the year gives but very little indication of the actual performance of the valve when on the car in service and that, inasmuch as your brake valve manipulation was accurate enough to release all of the triple valves to the rear and in front of this particular one, it is quite likely that this one would also have released if it had been in as good condition as the rest at the time the trouble was experienced.

Leak at Brake Valve Exhaust Port.

J. M. B., Columbus, O., writes: What causes a leak from the train line exhaust port of the H-6 automatic brake valve when the brake valve handle is on lap position?

A.—It is caused either by a leak from the equalizing reservoir pressure to the atmosphere or from a leak into the brake pipe. The difference can usually be told by observing the brake pipe and equalizing reservoir gage hands. If the equalizing reservoir gage hand falls it indicates leakage from the equalizing reservoir to the atmosphere but if the gage hands remain stationary it points to leakage into the brake pipe. These leaks may occur at any of the following points. A leak out of the equalizing reservoir may be
In the equalizing reservoir,
In the equalizing reservoir pipe,
In the equalizing reservoir gage pipe,
A leak in the tubes of the air gauge,
A leak on the seat of the rotary valve, leaking equalizing reservoir pressure to the atmosphere;
Leakage from the brake valve body gasket,
A leak from the equalizing reservoir pressure through a flaw in the brake valve casting, the latter usually found between the equalizing reservoir and release pipe port.

The leaks into the brake pipe that may be found at different times, are:
From the rotary valve seat of the automatic brake valve,
Through the brake valve body gaskets,
Through a flaw in the brake valve body,
Through the distributing valve body gasket,
Through a flaw in the distributing valve, or if the blow occurs only after brake-pipe pressure is lowered below the pressure in the brake cylinders,
Through the equalizing cylinder cap gasket, or, if the valve has a quick action cylinder cap
Through or past the seat of the brake cylinder check valve.

In addition to this, if the brake valve cutout cock is located in the reservoir pipe, the leak may be through the cutout cock, and with some systems of air operated water scoops, the leak into the brake pipe may be back through the check valve from the water scoop reservoir.

Wrongly Connected Air Pipes.

J. T., Baltimore, Md., writes: On an engine just out of the shop, with a new distributing valve and two overhauled brake valves, it was found that there was a constant blow from the emergency exhaust port of the automatic brake valve and both brake valves were taken apart and examined and also the distributing valve and nothing was found wrong. The blow at the exhaust port still continued and the engine was taken back to the shop and I did not find out what was wrong. Could you tell me what could cause this blow?

A.—It is quite likely that some of the small copper pipes were wrongly connected, and this is the first thing that should be looked for when there is any unusual action of the brake on an engine just out of the shop. Under such a condition it will usually be found that the reducing valve pipe is connected to the independent brake valve where the release pipe should be and in one case of this kind we can recall where the equalizing reservoir pipe was connected to the application cylinder passage of the distributing valve and the equalizing reservoir to the independent brake valve, either one of which will cause the blow at the emergency exhaust of the automatic brake valve.

Reboring Air Pump Bushings.

W. H. B., St. Louis, Mo., writes: Is there any device on the market for reboring a main valve bushing of an air pump without removing it from the head, and is it a labor saving device?

A.—We know there are devices of this kind on the market or can be made, and in the March, 1915, issue, page 78, a device for boring the main valve bushing without removing it from the head was partially illustrated with a description, but it is a question whether such tools are always a labor saving or economical installation for the reason that the average main valve bushing can be knocked out of the head with a maul in about two minutes' time and a good lathe hand can rebore three or four of them while one of these devices is being set up and the job started.

Babbitt Metal.

Perhaps the most common alloy producing what is known as babbitt metal consists of 24 pounds of lead, 2 pounds of tin, and 2 pounds of antimony.

the inner sheets, as indicated in the following table:

Mark No.	Location	Outside Wrapper Sheet Expansion
1	Side Sheet	3/16 inch
2	Side Sheet	5/32 inch
3	Side Sheet	1/4 inch
4	Side Sheet	3/16 inch
5	Side Sheet	7/32 inch
6	Wagon Top	7/32 inch
1	Throat Sheet	5/32 inch
2	Throat Sheet	3/16 inch
1	Back Head	5/32 inch

Location	Inside of Firebox Expansion	Difference in Expansion
Side Sheet	1/8 inch	1/16 inch
Side Sheet	3/32 inch	1/16 inch
Side Sheet	5/32 inch	3/32 inch
Side Sheet	1/8 inch	1/16 inch
Side Sheet	1/8 inch	3/32 inch
Crown Sheet	5/32 inch	1/16 inch
Tube Sheet	7/64 inch	3/64 inch
	1/8 inch	1/16 inch

(No Record on door sheet)

thus verifying our opinion in that regard, and seemingly accounting for the break-

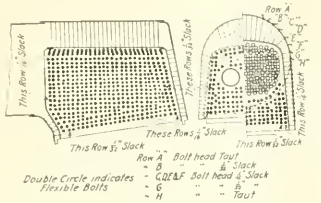


FIG. 6.

age of the back heads and throat sheets along the outer row of staybolts; also the vertical cracks in the side sheets, as well as the cracks from the water-bar holes, as indicated in Figs. 2, 3, 4 and 5. This theory is borne out by the movement of the needle shown extending through the throat sheet, Fig. 1, as, when the fire was first started, and before circulation was fully established, the needle moved out 3/32 inch, and later, when circulation was established and the steam pressure began to rise, the needle moved backward about 1/16 inch. The first movement of

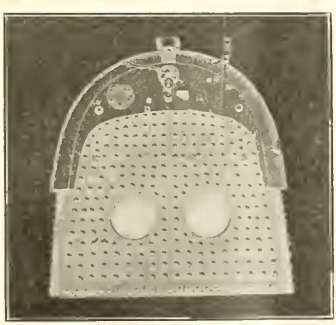


FIG. 7.

the needle, we think, throws some light on the cause of side sheets puffing along the fire line, as they sometimes do.

Fig. 6 shows, to the left, an installation of flexible staybolts, which, when adjusted, as shown by the tables and notes, in our opinion, will relieve the stresses that are the cause of the failures indicated in Figs. 2 and 3; and, to the right, the same installation and method of adjustment of flexible staybolts in the first and second rows on the side at each end to relieve the stresses that we believe are responsible for the failures shown in Figs. 4 and 5.

Our experience on the New York Central and Michigan Central has been of the most satisfactory nature in preventing cracked sheets and leaky side-sheet seams, and we are of the opinion that an installation of flexible staybolts, set loose, as shown in Fig. No. 6, with this difference, we believe a full installation in the throat sheet, set loose as follows, is advisable:

First row above mud rings..Tight
Second row above mud ring.1/32 in. loose
All others .....1/16 in. loose

And the back flue-sheet braces 3/32 in. loose and will increase the life of a modern firebox from 50 to 75 per cent.; the loose installation of flexible staybolts in the throat sheet, and slack braces to back flue sheet, that will be shown in Fig. 17, being considered necessary in order to avoid excessive staybolt and flue-sheet brace breakages, and at the same time reduce the strain on the arch-flue anchorages, the latter being quite a source of trouble when they begin to blow, especially at the front end.

Figs. 7 and 7a represent the first engine in the United States, I believe, that ever had an entire installation of flexible staybolts. It is an Atlantic type, and the full installation of flexible staybolts was made in February, 1907, when the engine received a new firebox. All staybolts were of the Tate flexible type with the exception of four bolts under the popper dome and four under the steam turret, and eight bolts which go on top seam of back head. Those that were not flexible were sling stays, as shown in Fig. 8.

In February, 1908, this engine again received general repairs, and the firebox was in good condition. All caps were removed from the flexible staybolts and they were found to be in good condition. In April, 1909, it was found necessary to patch the top of the back tube sheet on account of the sheet cracking from the tube hole around flange. This was done in the engine house. Also in that year, this engine again received general repairs, and by that time the top of back tube sheet had given out in two more places, but as sheet was in good condition otherwise it was again patched. Three of the mud ring corners also had to have small patches applied. No broken staybolts were found at that time. In April, 1911, the engine was again in shop for general repairs, at which time all caps were removed from the flexible staybolts and the staybolts were found to be sound. Patch was applied on top flange of flue sheet on left side.

This engine had made 319,067 miles to November 1, 1911. In July, 1912, the engine was again in shop for general repairs, and all flues were reset, mud ring corners calked, and new back tube sheet applied; crown sheet and water space cleaned; caps were removed from all flexible staybolts and no broken staybolts were found.

Up to January 1, 1914, this engine had made 440,189 miles dating from the time

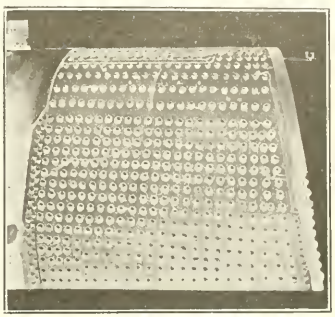


FIG. 7a.

new firebox was applied in February, 1907.

In March, 1914, engine was again in shop for general repairs,—caps were removed from flexible staybolts and all staybolts were good, a patch was applied on right side sheet, size 24 x 36; patches were also applied on corners both right and left sides. In March, 1915, when the engine was again in shop for machinery repairs, both front corners, also left back corner, were patched, and no broken staybolts were found at that time. In October, 1915, superheater was applied, also new side sheets, flue sheets and door sheet, and no broken staybolts were found. Up to that time this engine had made a total of 500,000 miles, without a flexible staybolt failure.

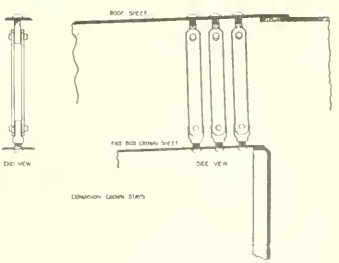


FIG. 8.

This is the same engine showing the installation as it was made in the sides. The manner of installing the bolts on that engine was probably not quite as methodical and up-to-date as we do it nowadays, but it was fairly good."